

Neuroscientific Frontiers — Module 06: Learning —

Study Questions

1. How do LTP and LTD implement parameter learning in the generative model? Map each plasticity mechanism to specific matrix updates (A, B, C, D).
2. What is spike-timing-dependent plasticity (STDP)? How does it implement causal inference at the synaptic level?
3. How do reward prediction errors (RPEs) update the C vector (preferences)? Trace the dopaminergic circuit from VTA to striatum to PFC.
4. What is the relationship between prediction error precision and learning rate? Why do high-precision errors produce faster learning?
5. Explain the complementary learning systems theory. How do hippocampal (fast) and neocortical (slow) learning differ in their computational roles?
6. What are sharp wave-ripples (SWRs)? How do they implement memory consolidation as model parameter transfer between brain systems?
7. How does the three-way coupling (SWR $\rightarrow$ spindle $\rightarrow$ slow oscillation) create a temporal window for cortical plasticity? Why is this coordination necessary?
8. How do sleep spindles facilitate LTP? Why do individual differences in spindle density predict learning ability?
9. What is Bayesian Model Reduction? How does synaptic pruning during development implement BMR — removing unnecessary model parameters?
10. How does the synaptic homeostasis hypothesis (Tononi & Cirelli) relate to Active Inference? Why does sleep improve generalization by implementing a form of BMR?
11. What is reconsolidation? How does memory reactivation create a window for model revision? What clinical implications does this have for PTSD treatment?
12. How might adult hippocampal neurogenesis implement structure learning — adding new nodes to the generative model? What is the target computational benefit?
13. How does environmental enrichment affect neurogenesis? Connect this to the idea that complex environments require richer model structure.
14. What is metamemory? How does the brain track the reliability of its own memory (the precision of recalled parameters)?

15. How does Active Inference explain the spacing effect (distributed practice > massed practice)? What changes occur at the synaptic level with spaced learning?
16. How does catastrophic forgetting in artificial neural networks relate to the complementary learning systems framework? How does the brain avoid catastrophic forgetting?
17. What is schema-consistent learning? How does having a strong prior model (schema) facilitate rapid learning of schema-consistent information?
18. How does Active Inference explain reconsolidation-triggered memory updating in extinction learning? Why is extinction not "forgetting"?
19. Compare STDP-based causal learning with Granger causality analysis used in DCM (Research Methods track). What is the relationship between synaptic-level causal inference and system-level causal inference?
20. Critically evaluate: Is the mapping between BMR and synaptic pruning a genuine mechanistic correspondence, or merely an analogy? What experimental evidence would strengthen or weaken this mapping?